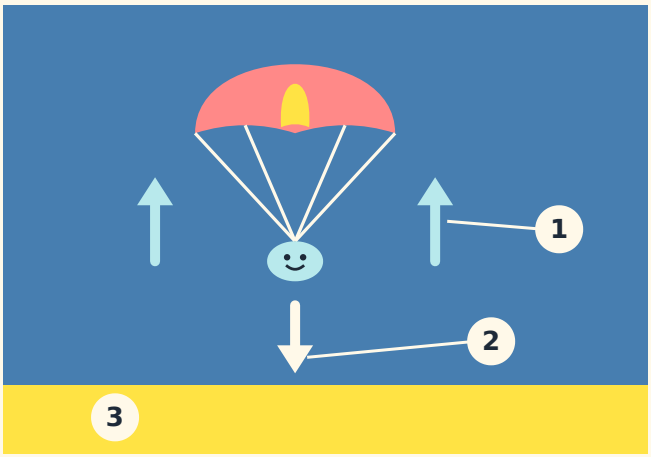
Gravity Detective Activity Sheet

Name: _____ Date: _____

1 Label the forces

Write the right word next to each number. Use the word bank to help you.



1 _____

2 _____

3 _____

Word bank: air resistance (pushes up) · gravity (pulls down) · Earth

2 The great drop race

Predict first: which will land first? Then drop them together and circle what happened.

The race	My prediction	What happened
Flat sheet v scrunched ball	flat / ball / same	flat / ball / same
Flat sheet v folded sheet	flat / folded / same	flat / folded / same
Small ball v big ball	small / big / same	small / big / same

Draw both papers falling. Show the air pushing on them.

Flat sheet

Scrunched ball

3 Pull back. Let go. Launch!

Fire the pom-pom at the target three times. Measure how far it went.

Try	How far did it go?	Did it hit the target? Circle.
Try 1	_____ cm	yes / no
Try 2	_____ cm	yes / no
Try 3	_____ cm	yes / no

4 Catapult challenge: record your findings

Will more pom-poms fly further or less far? **Circle one:** further / less far / the same

Load	Try 1	Try 2	Try 3
1 pom-pom	_____ cm	_____ cm	_____ cm
2 pom-poms	_____ cm	_____ cm	_____ cm
3 pom-poms	_____ cm	_____ cm	_____ cm

Did it match your prediction? **yes / no**

Which load landed closest? But why?

5 Which was your favourite experiment?

Circle one, then say why.



Drop race



Catapult



Catapult challenge

Because

What would you change next time?